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Crosscutting Areas

Regime-Dependent Approximations for the Single-Item Dynamic Pricing Problem

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Abstract. We study the single-item dynamic pricing problem in three separate asymptotic regimes characterized by their ratio of inventory to market size. We first consider the case of customer item valuations following an exponential distribution, for which we derive a sharp characterization of the boundaries between each of the regimes. We then proceed to the case of customer item valuations following a general distribution. In this case, we derive for each regime approximations to the optimal value function, optimal pricing policy, and optimal purchasing probability policy for both the offline and the online setting. We also provide for each regime an approximation to the regret of the optimal online value function relative to the optimal offline value function. In addition to these results, we show that in the regime when the inventory-to-market size ratio is low, a static run-out rate policy asymptotically fails to be first-order optimal. However, a dynamic run-out rate policy is shown to achieve both first- and second-order optimality. Finally, in the numerics section, we test the quality of our approximations and determine the boundaries between each of the three asymptotic regimes in the case of general customer item valuation distributions.

Supplemental Material: All supplemental materials, including the code, data, and files required to reproduce the results, are available at https://doi.org/10.1287/opre.2025.1656.

Keywords: dynamic pricing • regret analysis • stochastic optimization • optimal control • asymptotics • market size • inventory • extreme value theory • calculus of variations

1. Introduction

Dynamic pricing has become an essential operational strategy for firms seeking to maximize revenue through effective resource allocation. Successfully implementing dynamic pricing requires firms to continuously adjust prices based on current inventory levels and anticipated future demand. This intricate dependency creates a complex interplay between pricing decisions and inventory dynamics, making optimal pricing inherently challenging. In fact, closed-form solutions to the single-item dynamic pricing problem are known in only a few problem instances (Gallego and Van Ryzin 1994, McAfee and te Velde 2008, Gallego and Topaloglu 2019), highlighting the need for effective approximation methods and practical policies.

A widely used approach to approximate solutions for the single-item dynamic pricing problem is the fluid regime proposed by Gallego and Van Ryzin (1994), in which inventory and market size scale proportionally. Although this assumption is often reasonable, recent market developments, including the rise of online marketplaces, social media, and viral marketing, have allowed firms with limited inventory to quickly reach larger customer bases. However, in many such contexts, inventory levels cannot easily adjust to match rapid growth in market demand or sudden shifts in consumer interest due to inherent production limitations, logistical restrictions, or deliberate strategic decisions aimed at minimizing holding costs and maintaining product exclusivity. Alternatively, firms may also misjudge demand or experience a sudden shift in customer interest, resulting in inventory levels that exceed actual market needs. These diverse market conditions represent significant imbalances between available inventory and potential market size, and it is unclear whether the approximations and insights of the fluid regime still hold. Therefore, to gain a more comprehensive understanding of the single-item dynamic pricing problem, it is essential to examine settings where the imbalance between inventory and market size is more pronounced than that captured by the traditional fluid scaling assumption.

In this paper, we simultaneously study the single-item dynamic pricing problem across three separate

asymptotic regimes, each representing different market conditions characterized by the ratio of the firm's inventory level to its market size. Situations where the ratio is high are referred to as the small market regime. Moderate values of the ratio correspond to the classical fluid regime. The last regime where inventory levels are low relative to the market size is referred to as the large market regime. Whereas prior literature predominantly focuses on the fluid regime, our analysis directly addresses the small and large market regimes, where inventory-demand imbalances are more pronounced. Collectively, these three regimes provide a comprehensive regime-dependent framework for analyzing dynamic pricing across different market conditions.

At first glance, the optimal pricing policies in the small and large market regimes may appear trivial in the limit, corresponding to a constant monopoly price in the former and increasingly high, potentially infinite prices in the latter. However, focusing only on these limiting cases overlooks how the optimal policy and value function behave as inventory and market size become progressively imbalanced. By explicitly analyzing the convergence toward these limits, we uncover important structural insights not captured by traditional fluid approximations. In particular, this enables us to derive more precise approximations in scenarios where fluid-based approximations can be less accurate.

We begin our analysis by considering the case where customers value the item being sold according to an exponential distribution. The benefit of studying this case first is that there exists a closed-form expression for the optimal expected revenue of the firm. This explicit solution enables us to derive precise approximations to the optimal value function across all three asymptotic regimes. We demonstrate that these approximations correspond to three distinct regions within the parameter space of inventory levels and market size. Furthermore, we obtain asymptotic expressions for the boundaries separating these regions and demonstrate that they can be effectively characterized by the ratio of the firm's initial inventory to its market size. These results demonstrate that the appropriate approximations differ fundamentally across regimes and emphasize the need for a regime-specific approach, rather than relying on a single approximation regime.

We next proceed to the case of more general item valuation distributions. The main results of the paper provide asymptotic approximations to the optimal value function, pricing policy, and purchasing probability policy for the single-item dynamic pricing problem in each of the three asymptotic regimes. These results are obtained for both the offline and online versions of the problem. The results that we obtain for the offline problem are apparently new. The small market and fluid results in the online setting are previously known (Gallego and Van Ryzin 1994). The large market results in the online setting are new, and we demonstrate both theoretically and numerically that the large market regime occupies a significant portion of the problem parameter space. We also use our offline and online approximations in each regime to derive asymptotic expressions for the regret of the optimal policy in the online setting relative to the offline setting.

The results from the large market regime are new throughout the paper and merit further discussion. Under the assumption that customer valuation distributions belong to the Gumbel domain of attraction in extreme value theory, we derive asymptotic approximations for the optimal expected revenue, pricing policy, and purchasing probability policy in both offline and online settings. A notable finding is that the first-order approximation of the optimal expected online revenue in the large market regime differs from the classical fluid regime result of Gallego and Van Ryzin (1994). In contrast, the first-order approximations for optimal pricing and purchasing probability policies coincide structurally with a dynamic run-out rate policy, consistent with the resolving heuristic (continuous reoptimization of the deterministic fluid problem) known from the fluid regime literature (Maglaras and Meissner 2006, Jasin 2014). Interestingly, we also find that in the large market regime, the order of the remainder term in the optimal expected revenue and pricing policy approximations explicitly depends on the Mills ratio (reciprocal hazard rate) of the customer valuation distribution. Furthermore, we show that in the large market regime, a static run-out rate policy fails to achieve first-order optimality, whereas the dynamic run-out rate policy maintains both first-order and second-order optimality. This contrasts sharply with the fluid regime, where both static and dynamic run-out rate policies achieve first-order optimality.

The remainder of the paper is organized as follows. In Section 2, we conduct a literature review. In Section 3, we present the model which is used throughout the paper. Section 4 provides the details of the small market, fluid, and large market regimes. In Section 5, we consider the special situation of customer item valuations following an exponential distribution, in which case we asymptotically characterize the boundaries between the three regimes. In Section 6, we provide the details of the extreme value theory Gumbel domain of attraction. In Section 7, we study the offline version of the model presented in Section 3 and provide approximations to its optimal value function in all three regimes. Section 8 contains our results for the online problem presented in Section 3. In Subsection 8.1, we first analyze the performance of static pricing policies in the large market regime. Then, in

Subsection 8.2, we provide the asymptotics on the optimal expected revenue, pricing policy, and purchasing probability policy in the small market, fluid, and large market regimes. In Subsection 8.3, we study the asymptotics of the regret of the optimal expected revenue with respect to its offline upper bound in all three regimes. In Subsection 8.4, we show how to construct asymptotically optimal dynamic pricing policies in the large market regime using our results from Section 8.2. In Subsection 8.5, we explicitly derive approximations for two specific valuation distributions, illustrating the general results presented in the previous sections. Finally, in Section 9, we present the results of several numerical experiments. All proofs may be found in the Online Appendix.

2. Related Literature

The classical single-item dynamic pricing problem was originally studied by Gallego and Van Ryzin (1994), who proposed the fluid regime that scales both the customer arrival rate and initial inventory level. They showed that a static pricing policy is asymptotically optimal in this regime. There have since been many extensions of the classical model of Gallego and Van Ryzin (1994), which include multiple products (Gallego and Van Ryzin 1997, Kunnumkal and Topaloglu 2010, Chen et al. 2019, Lei and Jasin 2020), learning (Araman and Caldentey 2009, Besbes and Zeevi 2012, Keskin and Zeevi 2014, den Boer and Zwart 2015), consumer behavior (Liu and Cooper 2015, Chen and Farias 2018, Gao et al. 2022), and competition (Adida and Perakis 2010, Martínez-de Albéniz and Talluri 2011, Gallego and Hu 2014). For a thorough overview of the literature, one may refer to Bitran and Caldentey (2003), Elmaghraby and Keskinocak (2003), Talluri and Van Ryzin (2004), and, more recently, Gallego and Topaloglu (2019) and Lobel (2021).

A line of research initiated by Maglaras and Meissner (2006) and further developed by Chen and Farias (2013) and Jasin (2014) introduces and analyzes continuous resolving heuristics based on fluid approximations. These policies resolve the deterministic fluid problem by updating inventory and the remaining time horizon. In the context of multi-item dynamic pricing, Jasin (2014) shows that such a resolving approach, even when applied periodically, can achieve performance improvements over static policies and closely track the optimal value under certain assumptions. Nonetheless, there are some scenarios in which the resolving heuristic performs worse than a corresponding static one; see Cooper (2002), Secomandi (2008), and Jasin and Kumar (2013). Although our proposed dynamic run-out rate policy is essentially the continuous resolving heuristic in the single-item setting, it is derived and analyzed in a very different asymptotic regime. In particular, we establish the first- and second-order optimality in the large market regime, where the market size significantly outpaces the available inventory, and fluid-based performance guarantees no longer hold a priori. Our results, therefore, demonstrate the robustness of the fluid-based resolving heuristic beyond the fluid regime assumptions.

A growing body of literature also analyzes the regret for different classes of dynamic resource allocation problems. A notable contribution in this area is the work of Arlotto and Gurvich (2019), who propose an algorithm for the multisecretary problem that achieves a constant regret when customer types (valuations) are discrete. Vera and Banerjee (2021), Vera et al. (2021), and Bumpensanti and Wang (2020) extend these concepts to online packing, matching, and network management problems. However, constant regret policies may not hold with a continuum of types, as shown by Bray (2025) and Besbes et al. (2025). Balseiro et al. (2024) provide conditions that generalize the constant and logarithmic regret analysis. Wang and Wang (2022) also study the regret of the network dynamic pricing problem and demonstrate that the dynamic reoptimization heuristic based on the fluid regime policies achieves a constant regret under additional assumptions.

Another related stream of literature concerns prophet inequalities and, in particular, their random-order variants, often referred to as prophet secretary problems (Esfandiari et al. 2017), due to their close connection to the classical secretary problem. Whereas the secretary problem literature typically focuses on additive regret, the prophet inequality literature focuses on establishing bounds on the competitive ratio between the online policy and the offline optimum when selecting $q \geq 1$ out of t sequentially arriving random variables (in discrete time). For an overview of recent advances on prophet inequalities, see Correa et al. (2019). Most of the existing work on prophet inequalities analyzes static threshold policies for finite q (Arnosti and Ma 2023, Chawla et al. 2024), with exceptions such as Correa et al. (2017) and Jiang et al. (2025). To the best of our knowledge, these results are generally derived in terms of q alone and, hence, only yield asymptotic insights in a fluid regime where q and t are scaled together to infinity. A notable exception is Correa et al. (2021), who seem to be the first to coin the term *large market regime*. They maintain a fixed q while letting the total number of random variables t grow to infinity. Then, they apply extreme value theory to derive limiting competitive ratios for static threshold policies in this regime.

Our dynamic pricing problem parallels Correa et al. (2021) but differs along two key dimensions. First, we study and characterize the optimal adaptive dynamic pricing policies in continuous time, where the number of arrivals is a random variable. Second, we characterize the asymptotic behavior of the optimal (adaptive)

online policy and the offline solution. This joint characterization enables us to provide precise, distribution-dependent results on additive regret (and thus the competitive ratio), which may hold independent interest for the broader literature on prophet inequalities.

Although prior research on dynamic pricing, resolving heuristics, and regret minimization has primarily focused on fluid scaling, there is relatively limited understanding of how these insights extend to settings where inventory and demand scale asymmetrically. This paper addresses that gap by explicitly characterizing the optimal policy, value function, and (additive) regret structure across three distinct asymptotic regimes. Our results in the large market regime, in particular, provide rigorous theoretical foundations for heuristic policies widely applied in practice, which were previously understood only partially under the fluid regime assumptions. By utilizing extreme value theory and asymptotic analysis, we establish the robustness of the fluid-resolving heuristic policies, demonstrating that they remain effective and theoretically justified even under significant deviations from the classical proportional scaling assumption of the fluid regime. It is worth noting that den Boer (2023) implicitly investigates certain characteristics of the large market regime. However, their focus differs, examining only the probability of stocking out in a single-unit setting.

3. The Online Problem

We consider a firm selling a single item over a finite selling horizon of length $t > 0$. The initial inventory level of the item is q units, which we assume can take any value in $\mathbb{N} = \{0, 1, 2, \dots\}$. Customers arrive to purchase the item according to a Poisson process with a rate $\lambda > 0$ that is known to the firm. We assume that the nth customer to arrive has the idiosyncratic nonnegative valuation X_n for the item. Moreover, we assume that the X_n's are independent and identically distributed (i.i.d.) and denote their common distribution function by F, which we assume to have a finite mean. The firm has knowledge of the customer item valuation distribution F and the fact that the customer item valuations are i.i.d. but does not know their specific values. We refer to this setting as the online problem because the firm does not know in advance the arrival times of customers or their item valuations.

We denote the lower limit of the support of F by x_L and the upper limit by x_U, which may be equal to ∞. We also assume, for the sake of simplicity, that F has a positive density f on the entirety of its support. Finally, all random variables and processes are assumed to be defined on a common probability space $(\Omega, \mathcal{F}, P)$.

At each point in time $s \in [0, t]$, the firm sets a price for the item, which we denote by $p_s \in \mathbb{R}_+$. An arriving customer is willing to purchase the item if their surplus at the time of their arrival is nonnegative. That is, customer n is willing to purchase the item if X_n is greater than or equal to the price when they arrive. Thus, a customer arriving and encountering a price p has a probability $1 - F(p)$ of making a purchase. We also assume that for each customer, the marginal value of more than one unit of the item is negligible. Therefore, each customer will purchase at most one unit.

The firm may dynamically adjust the price of the item in response to the previous history of customer purchases and its current inventory level. Because of the Markovian nature of the system we have described, and for the sake of simplicity, we restrict attention to pricing policies which at each point in time are only dependent on the current level of inventory. Such policies can be shown to be optimal amongst the much larger class of predictable pricing policies (Brémaud 1981) but at the expense of additional machinery in our proofs. In the present paper, an $\mathbb{R}_+ \cup p_\infty$-valued pricing policy[1] $p = \{p_s, 0 \le s \le t\}$ on $(\Omega, \mathcal{F}, P)$ is said to be an admissible pricing policy if the following two conditions are satisfied.

I. For each $0 \le s \le t$, $p_s = p(\lambda; Q_s, t - s)$ where Q_s is the inventory level at time s.

II. The family of functions $\{p(\lambda; q, \cdot), q \in \mathbb{N}\}$ are measurable, with $p(\lambda; 0, \cdot) = p_\infty$.

We denote the set of admissible pricing policies by $\mathcal{V}$. Note that according to our definition, at each point in time, an admissible pricing policy depends not only on the remaining inventory level and length of the selling horizon, but also on the customer arrival rate.

Consistent with the revenue management literature (Gallego and Van Ryzin 1994), we assume that all items have zero marginal cost. Moreover, we assume the firm is risk-neutral and interested in maximizing its expected revenue. Given an admissible pricing policy $p \in \mathcal{V}$, we denote by $J_p(\lambda; q, t)$ the corresponding expected revenue of the firm over the selling horizon [0, t]. Now, taking the supremum of the expected revenue over all admissible pricing policies $p \in \mathcal{V}$, we obtain the optimal expected revenue of the firm given q units of initial inventory, which we denote by the optimal online value function

$$J^\star(\lambda; q, t) = \sup_{p \in \mathcal{V}} J_p(\lambda; q, t). \tag{1}$$

The Optimization Problem (1) may be viewed as a stochastic control problem. Specifically, applying the results of Brémaud (1981) and setting $\mathbb{N}_+ = \{1, 2, \dots\}$, we arrive at the following Hamilton-Jacobi-Bellman (HJB) equations characterizing the family $\{J^\star(\lambda; q, \cdot), q \in \mathbb{N}\}$ of optimal value functions.

Theorem 1. *The family of functions* $\{J^\star(\lambda;q,\cdot), q \in \mathbb{N}\}$ *are the unique solution to the system of equations*

$$\begin{aligned}\frac{\partial J^\star(\lambda;q,t)}{\partial t} &= \lambda \sup_{p\in\mathbb{R}_+}\{(1-F(p))(p-[J^\star(\lambda;q,t)\\ &\qquad - J^\star(\lambda;q-1,t)])\}, \quad t \geq 0,\ q \in \mathbb{N}_+,\\ J^\star(\lambda;q,0) &= 0, \quad q \in \mathbb{N}_+,\\ J^\star(\lambda;0,t) &= 0, \quad t \geq 0. \end{aligned} \tag{2}$$

Moreover, there exists an optimal solution $p^\star$ *to* (1) *such that for* $q \in \mathbb{N}_+$,

$$\begin{aligned} p^\star(\lambda;q,s) = \arg\max_{p\in\mathbb{R}_+}\{(1-F(p))(p-[J^\star(\lambda;q,s)\\ - J^\star(\lambda;q-1,s)])\},\ s \geq 0. \end{aligned} \tag{3}$$

The assumption that F has a finite mean implies that the supremum in (2) is finite as well. The existence of the arg max in (3) is then a consequence of the additional assumption that F is continuous. This, in turn, implies that there exists at least one optimal policy $p^\star$ in (1). That is, the supremum in (1) may be converted into an arg max.

We conclude this section by noting that because the item valuation distribution F has a positive density on its support, its inverse function $F^{-1}(\pi) = \{x \in [x_L, x_U] : F(x) = \pi\}$ is uniquely defined on [0, 1]. This implies, as is common in the literature (Gallego and Van Ryzin 1994, Maglaras and Meissner 2006), that we may equivalently consider the decision of the firm at each point in time as that of choosing optimal purchasing probabilities instead of prices. In particular, given a desired purchasing probability $\pi_s \in [0,1]$ at time $s \in [0,t]$, at time $s \in [0,t]$, the firm may set a corresponding price $p_s = F^{-1}(1-\pi_s)$. Also note that because the inverse of F is uniquely defined, it follows that for each optimal pricing policy $p^\star$, there uniquely corresponds an optimal purchasing probability policy $\pi^\star = 1 - F(p^\star)$. Similarly, to each optimal purchasing probability policy $\pi^\star$, there uniquely corresponds an optimal pricing policy $p^\star = F^{-1}(1-\pi^\star)$.

4. Regime Definitions and the Inventory-to-Market Size Ratio

Given a customer arrival rate $\lambda > 0$ and selling horizon of length $t > 0$, the quantity λt is the expected number of customer arrivals over the length of the selling horizon. This quantity may be viewed as a measure of the size of the market in which the firm is selling. We therefore refer to λt as the market size. It turns out by a change-of-variables in (2) that the optimal value function satisfies $J^\star(\lambda;q,t) = J^\star(1;q,\lambda t) = J^\star(\lambda t;q,1)$. Thus, the optimal value function is dependent only on the initial inventory level and market size (as opposed to the customer arrival rate, initial inventory level, and length of the selling horizon). We may therefore represent the optimal value function by a two-parameter function $J^\star(q,\lambda t)$ from which the three-parameter version may be recovered using the former relationship. For the sake of simplicity, we mostly refer to the two-parameter version.

In this paper, we study the Single-Item Dynamic Pricing Problem (1) under three different scaling regimes, which we refer to as the small market, fluid, and large market regimes. The core structure of our regime-dependent analysis rests on how the initial inventory level q compares to the market size λt.

- *Small market regime*: $q \to \infty$ while λt is fixed (abundant inventory).
- *Fluid regime*: $q = c\lambda t$ with $0 < c < \infty$ (moderate inventory-to-market size imbalances).
- *Large market regime:* q is fixed while $\lambda t \to \infty$ (scarce inventory).

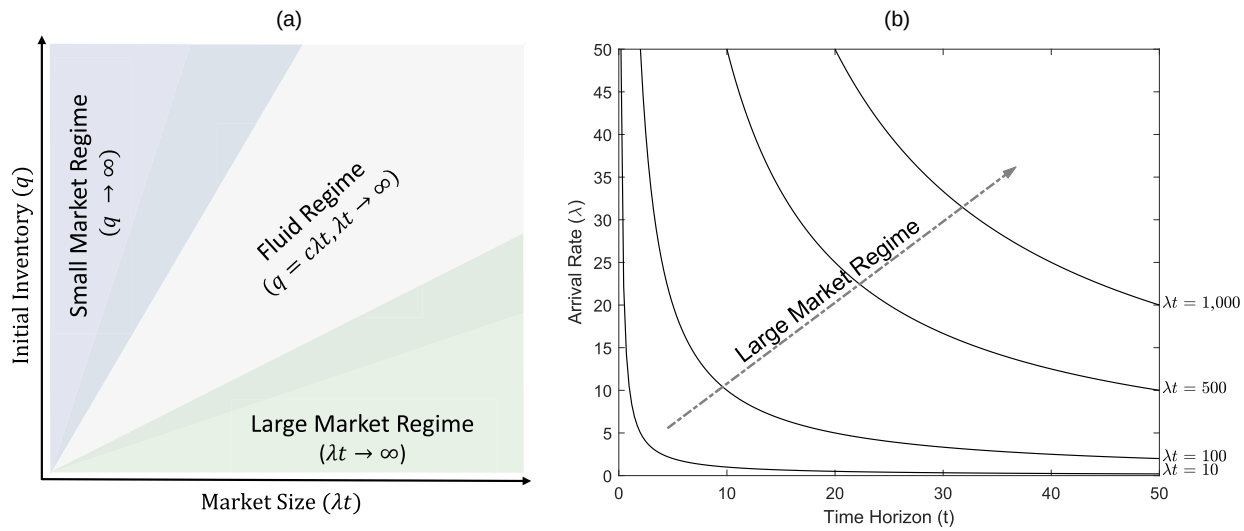
Figure 1(a) illustrates the three regimes within the parameter space defined by inventory level and market size. It is worth noting that the large market regime further decomposes into two distinct subregimes, depending on whether the market size grows by increasing the arrival rate λ or the time horizon t, as depicted in Figure 1(b). Specifically, holding λ and q fixed while letting $t \to \infty$ corresponds to scenarios where the firm has an extended selling period for a limited inventory. Conversely, holding t and q fixed while letting $\lambda \to \infty$ represents situations characterized by a high customer arrival rate within a fixed selling period, again with limited inventory. Mathematically, these two subregimes are equivalent.

It is important to note that the three scaling regimes are not just theoretical constructs, as they frequently occur in various practical situations. For instance, firms or sellers with limited production capacity or inventory but a broad online presence typically function within the large market regime. In contrast, companies with excess inventory but limited market exposure align more closely with the small market regime. Therefore, our framework effectively connects different market environments to their corresponding scaling regimes, providing a comprehensive understanding of the single-item dynamic pricing problem across various market conditions.

5. Exponentially Distributed Item Valuations

We first consider the special case in which customer valuations follow an exponential distribution. This case is analytically convenient because it admits explicit closed-form solutions for the optimal value function. Such tractability enables us to derive precise approximations for each of the three asymptotic regimes and clearly define the boundaries separating them. Importantly, this analysis clearly demonstrates how the

Figure 1. (Color online) (a) Relative Positioning of the Different Scaling Regimes and (b) Conceptual Representation of Scaling Within the Large Market Regime



accuracy of each regime-specific approximation depends on the inventory-to-market size ratio.

5.1. Optimal Value Function Approximations

For the sake of simplicity, in addition to assuming that the customer item valuation distribution is exponentially distributed, we assume that it has a mean of one. It then follows by Gallego and Van Ryzin (1994) that for each initial inventory level $q \in \mathbb{N}$ and market size $\lambda t \geq 0$, the optimal value function is given by

$$J^\star(q, \lambda t) = \ln\left(\sum_{k=0}^{q} \frac{(\lambda t/e)^k}{k!}\right). \tag{4}$$

Using some elementary properties of the incomplete gamma function, it is straightforward to derive approximations to the Optimal Value Function (4) in the small market, fluid, and large market regimes. Our three approximations are presented in Table 1. The details of their derivation may be found in the Online Appendix.

5.2. Regime Boundaries

We now ask the following question. Given a fixed initial inventory level $q > 0$ and market size $\lambda t > 0$, which of the three approximations in Table 1 provides the best estimate of the optimal value function $J^\star(q, \lambda t)$? We define *best* to be the approximation with the smallest distance in absolute value to the optimal value function. To begin to answer this question, for each $q > 0$ and $\lambda t > 0$, let $\text{Approx}^\star(q, \lambda t) = \text{S}$, F, or L, depending on whether the small market, fluid, or large market approximation, respectively, is optimal given q and λt. We then have the following result.

Table 1. Approximations to the Optimal Value Function for Each of the Three Asymptotic Regimes in the Case of Customer Item Valuations Following an Exponential Distribution with a Mean of One

Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
$\frac{\lambda t}{e}$	$q \ln\left(\frac{\lambda t}{q}\right)$	$q \ln\left(\left(\frac{(q/e)^q}{\Gamma(q+1)}\right)^{1/q} \cdot \frac{\lambda t}{q}\right)$

Proposition 1. *For each $q > 0$, there exists $0 < \sigma(q) < \tau(q)$ such that*

$$\text{Approx}^\star(q, \lambda t) = \begin{cases} \text{S} & \text{for } 0 < \lambda t < \sigma(q), \\ \text{F} & \text{for } \sigma(q) < \lambda t < \tau(q), \\ \text{L} & \text{for } \lambda t > \tau(q). \end{cases} \tag{5}$$

Moreover, $\sigma(q) < eq$ for $q > 0$, and $\tau(q) > eq$ for $q > 8/\pi \approx 2.55$.

Proposition 1 implies there exist three distinct regions of the $(q, \lambda t)$ parameter space corresponding to where each of the three approximations in Table 1 performs best. The function σ is the boundary between the small market and fluid regions, and τ is the boundary between the fluid and large market regions. There is no boundary between the small market and large market regions because the fluid region lies between them.

It is not possible to provide exact formulas for the boundaries σ and τ. However, letting $q \to \infty$, we can provide approximations to $\sigma(q)$ and $\tau(q)$. First, consider σ and recall (Olver et al. 2010) the definition of the error function

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-x^2} dx \text{ for } z \in \mathbb{R}, \tag{6}$$

and let erfc = 1 − erf denote the complementary error function. We then have the following.

Proposition 2. *The function σ has the asymptotic expansion*

$$\sigma(q) = qe - \gamma^\star e\sqrt{2q} + o(q^{1/2}) \text{ as } q \to \infty, \tag{7}$$

where $\gamma^\star > 0$ is the unique solution to

$$\frac{1}{2}\text{erfc}(-\gamma^\star) = \exp(-(\gamma^\star)^2/2). \tag{8}$$

We also have a similar result for the boundary τ.

Proposition 3. *The function τ has the asymptotic expansion*

$$\tau(q) = eq + \frac{e}{(2\pi)^{1/4}} \cdot q^{3/4} + o(q^{3/4}) \text{ as } q \to \infty. \tag{9}$$

5.3. Initial Inventory-to-Market Size Ratio

Propositions 2 and 3 imply that the boundaries between the three regimes can be approximated by the ratio of the initial inventory level to the market size. Thus, let

$$\text{IMR} = \frac{\text{initial inventory}}{\text{market size}},$$

denote the initial inventory-to-market size ratio. Intuitively, large IMR values correspond to the small market regime, whereas smaller values correspond to the large market regime. Values in between correspond to the fluid regime. Table 2 uses Propositions 2 and 3 to characterize the boundaries between the three regimes in terms of the IMR up to $o(q^{-1/2})$ and $o(q^{-1/4})$, respectively.

The small market regime corresponds to IMR values approximately greater than $1/e$, and the large market regime corresponds to IMR values approximately less than $1/e$. The fluid regime occupies a small interval of IMR values in a neighborhood of $1/e$. The width of this interval vanishes as either the initial inventory level q or the market size λt grows large. Thus, for most parameter combinations, either the small market or the large market approximation is the most accurate.

To illustrate this observation, we next plot the true and approximate boundaries in Figure 2. Figure 2 contains four plots of the approximations to $\sigma(q)$ and $\tau(q)$ derived from Propositions 2 and 3, neglecting the $o(q^{1/2})$ and $o(q^{3/4})$ remainder terms, respectively. Each plot captures a progressively larger portion of the $(\lambda t, q)$ parameter space. We also plot the true regime boundaries determined numerically using the Expression (4) for $J^\star(q, \lambda t)$. We first observe that the boundary approximations obtained from Propositions 2 and 3 are very close to their true values. Furthermore, as we extend the axis for larger values of initial inventories and market sizes, we observe that the space where the Fluid approximation is the best progressively diminishes, confirming the observation that for most parameter combinations, either the small market or the large market approximation is the most accurate.

Building on the boundary analysis, we now compare the magnitude of the approximation errors under the Fluid and Large Market regimes. Figure 3 presents a heatmap showing the difference in absolute relative errors from the optimal, computed as the absolute Fluid relative error minus the absolute Large Market relative error. Positive values indicate that the Large Market approximation is more accurate, whereas negative values favor the Fluid approximation. The Fluid regime outperforms near the Small Market-Fluid boundary, with differences reaching up to 40 percentage points in relative error. Conversely, the Large Market regime achieves a similar advantage in states with low inventory and large markets. Near the Fluid-Large Market boundary, the differences become negligible, indicating that the two approximations perform similarly in this region.

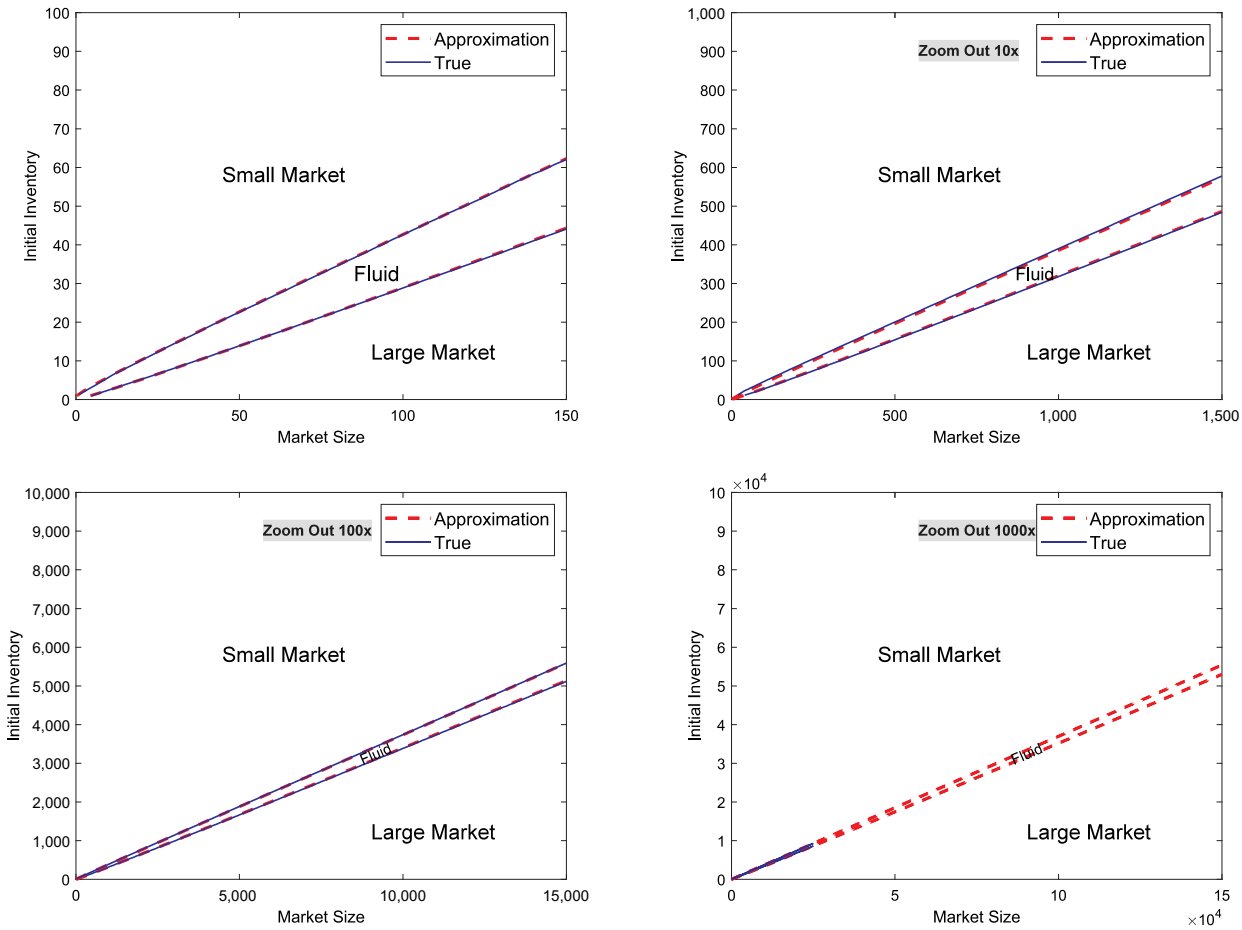
6. The Gumbel Domain of Attraction

Although the exponential valuation distribution offers useful boundaries for analysis, its applicability is somewhat restricted. To extend our analysis to a broader class of distributions, we leverage concepts from extreme value theory, which serves as the primary analytical tool for analyzing the large market regime. At a high level, extreme value theory describes the asymptotic behavior of the maximum of independent and identically distributed random variables, classifying distributions into three distinct *domains of attraction* based on their tail characteristics. In this paper, we specifically focus on distributions within the Gumbel domain of attraction. This domain is the largest among the three and encompasses commonly used valuation distributions, including exponential, normal, and lognormal, capturing a wide range of tail behaviors from light to moderately heavy. Moreover, distributions in the Gumbel domain may have either bounded or infinite support and always possess finite moments of all orders. A concise introduction to the key results and properties from extreme value theory essential to our analysis is provided in Online Appendix I. For a more comprehensive treatment, we refer readers to the books by Embrechts et al. (2013), Resnick (2013), and de Haan and Ferreira (2006).

Table 2. Approximate IMR Boundaries Between the Three Regimes

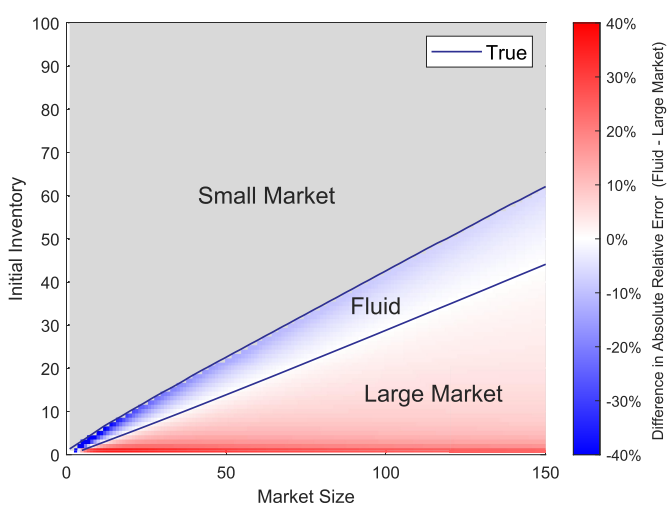
Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
$\text{IMR} > \frac{1}{e} + \frac{\gamma^\star}{e}\sqrt{\frac{2}{q}}$	$\frac{1}{e} + \frac{\gamma^\star}{e}\sqrt{\frac{2}{q}} > \text{IMR} > \frac{1}{e} - \frac{1}{e(2\pi q)^{1/4}}$	$\frac{1}{e} - \frac{1}{e(2\pi q)^{1/4}} > \text{IMR}$

Figure 2. (Color online) Boundaries Between the Three Regimes in the Case of Item Valuations with an Exponential Distribution with a Mean of One



For the small market and fluid regimes, our analysis requires minimal assumptions beyond those outlined in Section 3. In contrast, for the large market regime, we assume that valuations belong to the Gumbel domain of attraction and further restrict our attention to a subclass known as von-Mises functions. Importantly, this subclass assumption is not overly restrictive, as every distribution within the Gumbel domain

Figure 3. (Color online) Heatmap of the Difference in Absolute Relative Error to $J^{\star}(q, \lambda t)$ (Fluid Minus Large Market)



is tail-equivalent to a von-Mises function (Embrechts et al. 2013).

Definition 1 (von-Mises Function). An item valuation distribution F is said to be a von-Mises function if there exists an $x_0 < x_U$ and $c > 0$ such that for $x_0 < x < x_U$,

$$1 - F(x) = c \cdot \exp\left(-\int_{x_0}^{x} \frac{1}{u(t)} dt\right), \tag{10}$$

where $u(x) > 0$ for $x_0 < x < x_U$ and u is absolutely continuous on (x_0, x_U) with

$$u'(x) \to 0 \text{ as } x \to x_U. \tag{11}$$

The function u in (10) is sometimes (Resnick 2013) referred to as the auxiliary function associated with F. Differentiating both sides of (10), it is straightforward to see that $u(x) = (1 - F(x))/f(x)$ for $x \in (x_0, x_U)$, where f is the density of F. Thus, u is equal to the reciprocal hazard rate function or Mills ratio of F.

The von-Mises function assumption enables us to characterize the tail behavior of the valuation distribution directly by the Mills ratio $u(x)$ and its derivative. This structure ensures that our norming functions $a(t)$ and $b(t)$ behave regularly enough to support precise asymptotic expansions. In particular, if F is a von-Mises function, then we may define the norming functions b and a by setting

$$b(t) = F^{-1}(1 - 1/t) \text{ and } a(t) = u(b(t)) \quad \text{for } t > 1/(1 - F(x_0)). \tag{12}$$

Both functions b and a play a prominent role in the remainder of the paper. We also note (Resnick 2013) that $a(t) \in o(b(t))$ and that a is a slowly varying function.

In Subsection 8.5, we explicitly derive approximations for two specific valuation distributions, illustrating the general results presented in the subsequent sections.

7. Approximations for the Offline Problem

We now study an offline version of the model in Section 3. In the offline problem, the firm has full information on the number of customers who will arrive over the selling horizon, as well as each of their item valuations. That is, the sequence of random variables $\{X_n, n = 1, 2, \ldots, N_t\}$ is known to the firm. The objective of the firm is to find a pricing policy $p = \{p_s, 0 \le s \le t\}$ that is measurable with respect to the σ-algebra generated by $\{X_n, n = 1, 2, \ldots, N_t\}$ and maximizes the firm's expected revenue.

Although the offline problem is not of direct operational interest due to its assumption of complete foresight, it serves as a standard benchmark for defining regret in the online decision-making literature (see Subsection 8.3). Specifically, the offline value function provides a natural upper bound on achievable revenue, enabling the precise measurement of regret as the gap between offline-optimal performance and that of implementable online policies. The approximations we develop for the offline value function allow us to accurately quantify this regret and systematically evaluate the relative effectiveness of static and dynamic pricing policies in the large market regime.

Given $q \in \mathbb{N}_+$ units of initial inventory and a selling horizon of length $t > 0$, the revenue of the firm (in either the online or offline problem) is upper bounded by the $\min(q, N_t)$ highest item valuations from the sequence $\{X_1, X_2, \ldots, X_{N_t}\}$. Moreover, in the offline problem, the firm can achieve this upper bound by selling its items only to the customers with the top $\min(q, N_t)$ item valuations at a price exactly equal to each of their item valuations.

In the language of order statistics, the previous may be stated as follows. For each $n \in \mathbb{N}_+$ and $1 \le m \le n$, let $X_{m,n}$ denote the mth largest value in the sequence $\{X_1, X_2, \ldots, X_n\}$. The optimal offline value function of the firm is then given by

$$J^{\star}_{\text{OFF}}(\lambda; q, t) = E\left[\sum_{m=1}^{q \wedge N_t} X_{m,N_t}\right], \tag{13}$$

where we recall that N_t is a Poisson random variable with a mean of λt. It follows that the optimal offline value function is dependent only on the initial inventory level q and market size λt, and so we may represent it by a two-parameter function $J^{\star}_{\text{OFF}}(q, \lambda t)$.

In Proposition 11 of the Online Appendix, it is shown that the Optimal Offline Value Function (13) has the integral representation

$$J^{\star}_{\text{OFF}}(q, \lambda t) = \frac{1}{\Gamma(q)} \int_0^{\lambda t} F^{-1}\left(1 - \frac{v}{\lambda t}\right) \cdot \Gamma(q, v) \cdot dv, \tag{14}$$

for each $q \in \mathbb{N}_+$ and $\lambda t > 0$. This integral representation holds not just for item valuation distributions in the Gumbel domain of attraction, but for any item valuation distribution with a finite mean.

It also turns out that (14) may be used to construct approximations to the optimal offline value function. For each $q \in \mathbb{N}_+$, let

$$C_q = \sum_{m=1}^{q} (H_{m-1} - \gamma) = q \cdot (H_q - \gamma - 1), \tag{15}$$

where H_{m-1} is the $(m-1)$st harmonic number with $H_0 = 0$, and $\gamma \approx 0.57722$ is the Euler-Mascheroni constant. Table 3 provides three approximations to the optimal offline value function corresponding to the small market, fluid, and large market regimes. The derivations of these approximations may be found in

Table 3. Approximations to the Optimal Offline Value Function in Each of the Three Asymptotic Regimes

Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
$\lambda t E[X]$	$\lambda t \int_0^{q/\lambda t} F^{-1}(1-s)ds$	$qF^{-1}\left(1-\frac{e^{Cq/q}}{\lambda t}\right)$

Section B of the Online Appendix. The small market and fluid approximations hold for arbitrary item valuation distributions with a finite mean. The large market approximation is specific to item valuation distributions in the Gumbel domain of attraction satisfying the von-Mises Conditions (10) and (11).

8. Approximations for the Online Problem

We now proceed to study the online model of Section 3. In Subsection 8.1, we study the optimal static pricing policy for the online model in the large market regime. Subsection 8.2 contains our approximations to the optimal online value function, pricing policy, and purchasing probability policy, where we optimize over the set of all admissible pricing policies. In Subsection 8.3, we provide approximations to the regret of the optimal online value function relative to the optimal offline value function. Finally, in Subsection 8.4, we discuss the notion of asymptotic optimality for the online problem in the large market regime.

8.1. Static Pricing Policies

In this subsection, we study the class of static pricing policies in the large market regime. These policies set a single price at the outset of the selling horizon without any knowledge of the number of customers who will arrive or their specific item valuations, making them simple and easy to implement. Moreover, their optimal value function provides a lower bound for the optimal online value function.

Suppose at time $t = 0$ the firm sets a single price $p \in \mathbb{R}_+$ for the entire selling horizon. Then, given $q \in \mathbb{N}_+$ units of initial inventory and a selling horizon of length $t \geq 0$, the realized revenue of the firm is given in terms of the order statistics of the customer item valuations as

$$p \sum_{m=1}^{q \wedge N_t} 1\{X_{m,N_t} > p\}. \tag{16}$$

Taking the expectation of the above and recalling that N_t is a Poisson random variable with a mean of λt, the static pricing value function of the firm is given by

$$J^p_{\mathrm{STAT}}(q, \lambda t) = pE\left[\sum_{m=1}^{q \wedge N_t} 1\{X_{m,N_t} > p\}\right]. \tag{17}$$

The optimal static pricing value function is then

$$J^\star_{\mathrm{STAT}}(q, \lambda t) = \sup_{p \in \mathbb{R}_+} J^p_{\mathrm{STAT}}(q, \lambda t), \tag{18}$$

and the optimal static price is

$$p^\star_{\mathrm{STAT}}(q, \lambda t) = \arg\max_{p \in \mathbb{R}_+} J^p_{\mathrm{STAT}}(q, \lambda t). \tag{19}$$

One may interpret $p^\star_{\mathrm{STAT}}(q, \lambda t)$ as the optimal static price given an initial inventory level q and market size λt.

The first-order asymptotics of the optimal static pricing value function and the optimal static price in the large market regime are as follows. Recall from (12) the definition of the norming function b.

Proposition 4. *If F is in the Gumbel domain of attraction and satisfies the von-Mises Conditions (10) and (11), then for each $q \in \mathbb{N}_+$,*

$$\lim_{\lambda t \to \infty} \frac{J^\star_{\mathrm{STAT}}(q, \lambda t)}{b(\lambda t)} = q \text{ and } \lim_{\lambda t \to \infty} \frac{p^\star_{\mathrm{STAT}}(q, \lambda t)}{b(\lambda t)} = 1. \tag{20}$$

Now, recall (Resnick 2013) that b is a slowly varying function. It then follows by Proposition 14 in the Online Appendix that

$$\lim_{\lambda t \to \infty} \frac{J^\star_{\mathrm{STAT}}(q, \lambda t)}{J^\star_{\mathrm{OFF}}(q, \lambda t)} = 1. \tag{21}$$

It is therefore the case that on a first-order basis, the optimal expected revenues of the static pricing problem and offline problem are asymptotically the same in the large market regime. However, their second-order terms differ significantly, as we show next.

For each static price $p \in \mathbb{R}_+$, initial inventory level $q \in \mathbb{N}_+$, and market size $\lambda t > 0$, it is straightforward to see that the following inequality holds,

$$J^p_{\mathrm{STAT}}(q, \lambda t) \leq J^\star_{\mathrm{OFF}}(q, \lambda t). \tag{22}$$

It then follows that for each static price p, we may define its regret relative to the optimal offline pricing policy by setting

$$\mathrm{Regret}^p_{\mathrm{STAT}}(q, \lambda t) = J^\star_{\mathrm{OFF}}(q, \lambda t) - J^p_{\mathrm{STAT}}(q, \lambda t) \geq 0. \tag{23}$$

The minimum static pricing regret is obtained by maximizing the static pricing value function in (23) over all prices p. We denote the resulting quantity by

$$\mathrm{Regret}^\star_{\mathrm{STAT}}(q, \lambda t) = J^\star_{\mathrm{OFF}}(q, \lambda t) - J^\star_{\mathrm{STAT}}(q, \lambda t).$$

The following is our main result on the minimum static pricing regret in the large market regime. It is implied by Propositions 14 and 15 in the Online Appendix together with (A87). Recall from (12) the definition of the norming function a.

Proposition 5. *If F is in the Gumbel domain of attraction and satisfies the von-Mises Conditions (10) and (11), then for each $q \in \mathbb{N}_+$,*

$$\lim_{\lambda t \to \infty} \frac{\mathrm{Regret}^\star_{\mathrm{STAT}}(q, \lambda t)}{a(\lambda t)} = \infty. \tag{24}$$

This result implies that on a scale of $a(\lambda t)$, the optimal expected revenue of the offline problem is asymptotically infinitely larger than that of the static pricing problem in the large market regime. Recall also that the function a may be interpreted as the reciprocal of the hazard rate function of the customer item valuation distribution F. Thus, if $x_U = \infty$, then a may diverge to ∞, in which case, by the above, the static pricing regret itself will diverge, too.

The minimal regret in the large market regime can be significantly reduced by using dynamic pricing policies. See, for instance, Subsection 8.4 below, where the family of dynamic run-out rate policies is shown to be asymptotically optimal. It turns out, however, that static run-out rate policies do not perform as well. For each initial inventory level $q \in \mathbb{N}_+$ and market size $\lambda t > 0$, denote by $J^{\text{RUN}}_{\text{STAT}}(q, \lambda t)$ the value function associated with the static run-out rate price $F^{-1}(1 - q/\lambda t)$. We then have the following.

Proposition 6. *If F is in the Gumbel domain of attraction and satisfies the von-Mises Conditions* (10) *and* (11), *then for each $q \in \mathbb{N}_+$,*

$$\lim_{\lambda t \to \infty} \frac{J^{\text{RUN}}_{\text{STAT}}(q, \lambda t)}{b(\lambda t)} = q\left(1 - \frac{(q/e)^q}{q!}\right). \tag{25}$$

Proposition 6 follows from theorem 3 of Gallego and Van Ryzin (1994) together with the fact that the norming function b is slowly varying. It is straightforward to see that the right-hand side of (25) is strictly less than q. It therefore follows by Proposition 4 that in the large market regime, the value function associated with a static run-out rate policy is not first-order optimal. This result is contrary to the case in the fluid regime, where the value function associated with a static run-out rate policy does achieve first-order optimality (Gallego and Van Ryzin 1994).

We also note that in the large market regime, a static run-out rate policy asymptotically results in a higher price than the optimal static price. In particular, by Proposition 15 in the Online Appendix, for each $q \in \mathbb{N}_+$,

$$\frac{F^{-1}(1 - q/\lambda t) - p^\star_{\text{STAT}}(q, \lambda t)}{a(\lambda t)} \to \infty \text{ as } \lambda t \to \infty. \tag{26}$$

8.2. Dynamic Pricing Policies

We now consider the entire class of admissible pricing policies in the online setting of Section 3. Recall that for each $q \in \mathbb{N}_+$ and $\lambda t > 0$, the optimal value function satisfies $J^\star(\lambda; q, t) = J^\star(1; q, \lambda t) = J^\star(\lambda t; q, 1)$. It then follows from (3) that we also have $p^\star(\lambda; q, t) = p^\star(1; q, \lambda t) = p^\star(\lambda t; q, 1)$ and $\pi^\star(\lambda; q, t) = \pi^\star(1; q, \lambda t) = \pi^\star(\lambda t; q, 1)$. In other words, at each point in time, the optimal price and optimal purchasing probability depend only on the current inventory level and remaining market size. We therefore may represent these quantities by the two-parameter functions $p^\star(q, \lambda t)$ and $\pi^\star(q, \lambda t)$, respectively.

The following result provides a sharp approximation to the optimal value function $J^\star(q, \lambda t)$, pricing policy $p^\star(q, \lambda t)$, and purchasing probability $\pi^\star(q, \lambda t)$ in the large market regime. Notably, it shows that the optimal pricing policy mirrors the dynamic run-out rate structure from the fluid regime, but with a distribution-dependent remainder term.

Theorem 2. *If F is in the Gumbel domain of attraction and satisfies the von-Mises Conditions* (10) *and* (11), *then for each $q \in \mathbb{N}_+$ as $\lambda t \to \infty$,*

$$J^\star(q, \lambda t) = qF^{-1}\left(1 - \left(\frac{q!}{(q/e)^q}\right)^{1/q} \frac{q}{\lambda t}\right) + o(a(\lambda t)), \tag{27}$$

and

$$p^\star(q, \lambda t) = F^{-1}\left(1 - \frac{q}{\lambda t}\right) + o(a(\lambda t)), \tag{28}$$

and

$$\pi^\star(q, \lambda t) = \frac{q}{\lambda t} + o(1/\lambda t). \tag{29}$$

The results of Theorem 2 are new to the literature. In (27), the first-order term for the optimal online value function in the large market regime differs from $qF^{-1}(1 - q/\lambda t)$, which is the first-order term for the optimal online value function in the fluid regime of Gallego and Van Ryzin (1994).

Next, note that the remainder term in (27) is $o(a(\lambda t))$, where $a(\lambda t) = u(b(\lambda t))$ with u being the Mills ratio of the item valuation distribution F and $b(\lambda t) = F^{-1}(1 - 1/\lambda t)$. The implications of this for the Approximation (27) are as follows. If the hazard rate function of F converges to a positive constant as $x \to x_U$, then the function $a(\lambda t)$ in the remainder term in (27) converges to the reciprocal of this constant as $\lambda t \to \infty$, and so the remainder term itself vanishes. More generally, the function $a(\lambda t)$ in (27) is decreasing if the hazard rate function of F is increasing, in which case the remainder term vanishes, too. If the hazard rate function of F is decreasing, then $a(\lambda t)$ is increasing as $\lambda t \to \infty$. In the particular case in which the support of F is bounded, then under the assumptions of Theorem 2, it is guaranteed (Resnick 2013) that the hazard rate function diverges to ∞ as $x \to x_U$, and so the remainder term in (27) vanishes as $\lambda t \to \infty$. In all cases, $a(\lambda t)$ is a slowly varying function that is $o(b(\lambda t))$ (Resnick 2013).

Finally, by (28), if F is a von-Mises function in the Gumbel domain of attraction, then up to $o(a(\lambda t))$, the optimal pricing policy in the large market regime follows the same run-out rate form as in the fluid regime of Gallego and Van Ryzin (1994). By (29), the optimal purchasing probabilities are up to $o(1/\lambda t)$ of a run-out rate form as well. In Section 9, we test the accuracy of

Table 4. Approximations to the Optimal Online Value Function in Each of the Three Asymptotic Regimes

Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
$\lambda t p^\star(1 - F(p^\star))$	$qF^{-1}\left(1 - \frac{q}{\lambda t}\right)$	$qF^{-1}\left(1 - \left(\frac{q!}{(q/e)^q}\right)^{1/q} \frac{q}{\lambda t}\right)$

our approximations by running several numerical experiments.

Now, let

$$p^\star = \arg\max_{p \in \mathbb{R}_+} p(1 - F(p)), \tag{30}$$

denote the optimal monopoly price. Assuming that $r(p) = p(1 - F(p))$ is unimodal on $\mathbb{R}_+$, it follows that $p^\star$ is unique. Then, combining Theorem 2 together with Gallego and Van Ryzin (1994), we obtain Table 4, which provides approximations to the optimal value function in each of the small market, fluid, and large market regimes.

8.3. Regret Analysis

We now study the regret of the optimal online value function $J^\star(q, \lambda t)$ relative to its upper bound given by the optimal offline value function $J^\star_{\text{OFF}}(q, \lambda t)$. First, note that for each admissible online pricing policy $p \in \mathcal{V}$, customer arrival rate $\lambda > 0$, initial inventory level $q \in \mathbb{N}$, and selling horizon $t > 0$, we have the inequality

$$J_p(\lambda; q, t) \le J^\star_{\text{OFF}}(q, \lambda t). \tag{31}$$

That is, the value function associated with each admissible online pricing policy is no greater than the optimal offline value function. Therefore, for each admissible online pricing policy $p \in \mathcal{V}$, we may define its regret relative to the optimal offline policy by setting

$$\text{Regret}_p(\lambda; q, t) = J^\star_{\text{OFF}}(q, \lambda t) - J_p(\lambda; q, t) \ge 0. \tag{32}$$

It follows that maximizing the value function in (1) is equivalent to minimizing the regret in (32). The minimal regret of the online problem relative to the offline problem may thus be represented as a function of the initial inventory level and market size. Specifically,

$$\text{Regret}^\star(q, \lambda t) = J^\star_{\text{OFF}}(q, \lambda t) - J^\star(q, \lambda t). \tag{33}$$

Table 5 provides approximations to the minimal regret in each of the three asymptotic regimes. These approximations follow from Tables 3 and 4 and are derived in Online Appendix G. Note that the approximation to the minimal regret in the small market regime is linear in the market size λt and constant with respect to the initial inventory level q. On the other hand, the approximation to the minimal regret in the fluid regime is the product of the market size λt and a function of $c = q/\lambda t$. This can be seen by rewriting the approximation as $\lambda t \cdot c \cdot E[X - p(c) | X > p(c)]$. Finally, the approximation to the minimal regret in the large market regime is the product of a function of the market size λt and a function of the initial inventory level q. We therefore see that, for each regime, the approximation to the minimal regret takes a product form.

Table 5. Approximations to the Minimal Regret in Each of the Three Asymptotic Regimes

Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
$\lambda t(E[X] - p^\star(1 - F(p^\star))$	$qE[X - p(c) \mid X > p(c)]$	$\ln\left(\frac{e^q q!}{e^{C_q}}\right) a(\lambda t)$

We complete this section by analyzing the minimal regret for the large market regime in more detail. First, recall from Subsection 8.1 that the minimal static pricing regret in the large market regime is equal to ∞ on the scale of $a(\lambda t)$ as $\lambda t \to \infty$. This contrasts with the result in Table 5, which implicitly demonstrates that the minimal dynamic pricing regret in the large market regime is on the scale of $o(a(\lambda t))$ as $\lambda t \to \infty$ (see (A149) in Online Appendix G). Dynamic pricing, therefore, offers a significant improvement over static pricing in large market regimes.

Next, recall that $a(\lambda t) = u(b(\lambda t))$, where u is the reciprocal of the hazard rate function of the customer item valuation distribution F. Because minimal dynamic pricing regret in the large market regime is on the scale of $o(a(\lambda t))$, then, for a fixed initial inventory level $q \in \mathbb{N}_+$ and as the market size $\lambda t \to \infty$, the first-order term of the minimal regret is decreasing if the hazard rate function of the customer item valuation distribution is increasing. Conversely, the first-order term in the minimal regret is increasing if the hazard rate function of the customer item valuation distribution is decreasing. In the case when the hazard rate function of F converges to a constant, the minimal regret converges to the reciprocal of that constant multiplied by $c(q)$.

Finally, regarding the multiplier $\ln((e^q q!)/e^{C_q})$, using Lemma 3 and some elementary asymptotics of the harmonic numbers, it may be shown that

$$C_q = q \ln q - q + \frac{1}{2} + o(1) \text{ as } q \to \infty. \tag{34}$$

By (34) and Stirling's approximation (Olver et al. 2010), it then follows that

$$\ln\left(\frac{e^q q!}{e^{C_q}}\right) = q + \ln\sqrt{2\pi q} - \frac{1}{2} + o(1) \text{ as } q \to \infty. \tag{35}$$

8.4. Asymptotic Optimality

Theorem 2 of Subsection 8.2 provides asymptotics for the optimal online value function, pricing policy, and purchasing probability policy in the large market regime. However, for most customer item valuation distributions, the optimal pricing policy is still unknown. In this subsection, we construct a simple-to-implement pricing policy whose value function achieves the same asymptotics in the large market regime as the optimal value function in Theorem 2.

Recall from Subsection 8.3 that for each admissible online pricing policy $p \in \mathcal{V}$, its regret relative to the optimal offline value function is defined for each initial inventory level $q \in \mathbb{N}_+$ and selling horizon of length $t > 0$ by

$$\text{Regret}_p(\lambda; q, t) = J^\star_{\text{OFF}}(q, \lambda t) - J_p(\lambda; q, t) \geq 0. \tag{36}$$

The following is then our definition of an asymptotically optimal pricing policy in the large market regime.

Definition 2. An admissible pricing policy $p \in \mathcal{V}$ is said to be asymptotically optimal in the large market regime if for each $q \in \mathbb{N}_+$,

$$\lim_{\lambda t \to \infty} \frac{\text{Regret}_p(\lambda; q, t)}{a(\lambda t)} = c(q), \tag{37}$$

where $c(q)$ is given by (35) and the limit above holds for any sequence of (λ, t) such that $\lambda t \to \infty$.

The results of Subsection 8.3 imply that an admissible pricing policy $p \in \mathcal{V}$ is asymptotically optimal in the large market regime if on a scale of $a(\lambda t)$, it achieves the same regret with respect to the optimal offline value function as does the optimal pricing policy $p^\star$. Using the Relationship (33) between the minimal regret and the optimal online value function, another way to state this is to say that $p \in \mathcal{V}$ is asymptotically optimal if

$$\lim_{\lambda t \to \infty} \frac{J^\star(q, \lambda t) - J_p(\lambda; q, t)}{a(\lambda t)} = 0, \tag{38}$$

for any sequence of (λ, t) such that $\lambda t \to \infty$.

Now, consider the pricing policy $p \in \mathcal{V}$ such that for each $\lambda > 0$ and $q \in \mathbb{N}_+$ and $t > 0$,

$$p(\lambda; q, t) = F^{-1}\left(1 - \frac{q}{\lambda t}\right) \text{ for } \lambda t > q, \tag{39}$$

and $p(\lambda; q, t) = p^\star$ otherwise, where $p^\star$ is the optimal monopoly price given by (30). The Pricing Policy (39) is a dynamic run-out rate policy that is obtained by continuously resolving the fluid approximation of Gallego and Van Ryzin (1994) under updated inventory and time-to-go. This policy was first shown by Maglaras and Meissner (2006) to outperform static pricing heuristics numerically and later rigorously analyzed by Jasin (2014), who established that dynamic reoptimization achieves lower regret in the classical fluid regime. Theorem 3 shows that this performance extends far beyond the fluid setting and into the large market regime.

Theorem 3. *The dynamic run-out rate policy $p \in \mathcal{V}$ given by (39) is asymptotically optimal in the large market regime.*

Theorem 3 highlights the robustness of the dynamic run-out rate policy across both the fluid and large market regimes. This stands in contrast to the static run-out rate, which fails to be even first-order optimal (Proposition 6), and to the optimal static pricing policy, which is first-order optimal in the large market regime (Proposition 4) but fails to be second-order optimal (Proposition 5).

To further illustrate this point, we consider valuations drawn from an exponential distribution with rate one. Figure 4 plots the relative revenue gap, with respect to the optimal dynamic policy, for each of the three policies: static run-out rate, optimal static policy, and dynamic run-out rate. The comparison is shown in Figure 4 for an initial inventory $q = 1$ as the market size increases.

As the market size increases, Figure 4 shows that only the dynamic run-out rate continues to closely track the optimal value. The gap under the optimal

Figure 4. (Color online) Relative Revenue Gaps with Respect to the Optimal Dynamic Policy for $q = 1$ as a Function of Market Size in the Case of Item Valuations with an Exponential Distribution with a Mean of One

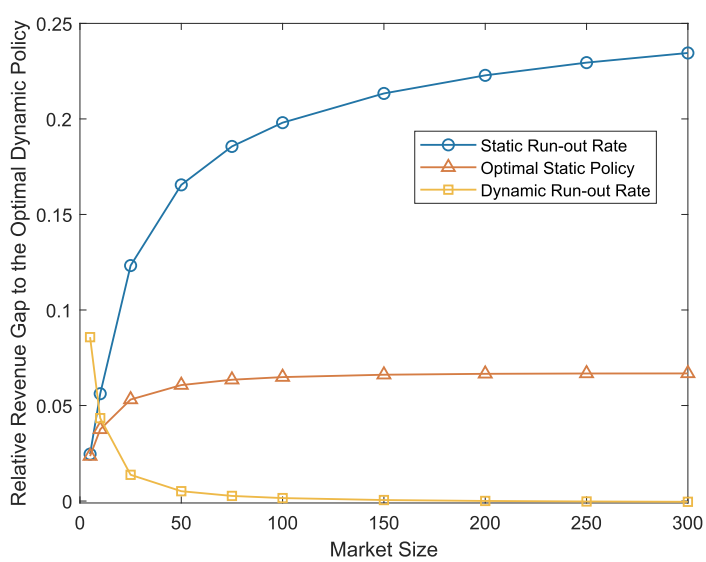


Table 6. Approximate Expressions Across Market Regimes Under a Finite-Endpoint Customer Item Valuation Distribution

Approx. type	Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
Optimal offline value	$\lambda t(x_U - (-\alpha\beta \cdot \mathrm{li}(1/\beta)))$	$q\left(x_U - \frac{\lambda t}{q}\left(-\alpha\beta \cdot \mathrm{li}\left(\frac{q}{\lambda t\beta}\right)\right)\right)$	$q\left(x_U - \alpha \ln^{-1}\left(\frac{\beta\lambda t}{e^{C_q/q}}\right)\right)$
Optimal online value	$\lambda t p_\alpha^\star(1 - F(p_\alpha^\star))$	$q\left(x_U - \alpha \ln^{-1}\left(\beta \cdot \frac{\lambda t}{q}\right)\right)$	$q\left(x_U - \alpha \ln^{-1}\left(\beta \cdot \left(\frac{(q/e)^q}{q!}\right)^{1/q} \cdot \frac{\lambda t}{q}\right)\right)$
Optimal price	$p_\alpha^\star = x_U - \frac{\alpha}{2}\left(\sqrt{1 + \frac{4x_U}{\alpha}} - 1\right)$	$x_U - \alpha \ln^{-1}\left(\beta \frac{\lambda t}{q}\right)$	$x_U - \alpha \ln^{-1}\left(\beta \frac{\lambda t}{q}\right)$
Minimal regret	$\lambda t(x_U + \alpha\beta \cdot \mathrm{li}(1/\beta) - p_\alpha^\star(1 - F(p_\alpha^\star)))$	$q\alpha\left(\ln^{-1}\left(\beta \cdot \frac{\lambda t}{q}\right) + \beta \cdot \frac{\lambda t}{q} \cdot \mathrm{li}\left(\frac{q}{\lambda t\beta}\right)\right)$	$\alpha \ln\left(\frac{e^q q!}{e^{C_q}}\right) \ln^{-1}\left(\frac{\beta\lambda t}{q}\right)$

static policy remains bounded away from zero, whereas the static run-out rate diverges sharply, especially in the large market regime. In Section 9, we provide a more extensive numerical evaluation of the Dynamic Run-out Rate Policy (39).

8.5. Examples

We now illustrate the general asymptotic results developed in the previous sections by explicitly working through two specific examples of valuation distributions. We derive the asymptotic approximations for the optimal offline and online value functions, as well as the optimal pricing policies and minimal regret for each example. The first example examines valuation distributions with finite upper endpoints, representing scenarios in which customer valuations are inherently bounded. The second example analyzes the Weibull distribution, which offers a flexible tail structure. These examples demonstrate clearly how the general regime-dependent approximations specialize under explicit distributional assumptions and highlight their implications for optimal policies and regret.

Our first example considers a bounded support distribution in the Gumbel domain of attraction (Embrechts et al. 2013). However, it is worth noting that not all bounded support distributions belong to the Gumbel domain of attraction.

Example 1 (Finite Right Endpoint). The tail of the distribution is given by

$$1 - F(x) = \beta \exp\left(-\frac{\alpha}{x_U - x}\right) \text{ for } 0 \le x < x_U, \tag{40}$$

where $\alpha > 0$ and $\beta = \exp(\alpha/x_U)$. In this case,

$$F^{-1}(p) = x_U - \frac{\alpha}{\ln(\beta/(1-p))} \quad \text{for } 0 \le p < 1. \tag{41}$$

For this distribution, the Normalizing and Centering Functions (12) are given by

$$b(\lambda t) = x_U - \frac{\alpha}{\ln(\beta\lambda t)} \quad \text{and} \quad a(\lambda t) = \frac{\alpha}{\ln^2(\beta\lambda t)}. \tag{42}$$

The explicit approximations for the Finite Right Endpoint distribution are provided in Table 6.

Our second example is the Weibull distribution, which can be viewed as a generalization of the exponential distribution. The Weibull distribution has an unbounded support and exhibits flexible tail behavior, as its parameters determine whether it has an increasing, constant, or decreasing hazard rate.

Example 2 (Weibull Distribution). The tail of a Weibull distribution is given by

$$1 - F(x) = \exp(-(x/\mu)^k) \text{ for } x \ge 0, \tag{43}$$

where $\mu > 0$ is a scale parameter and $k > 0$ is a shape parameter. In this case,

$$F^{-1}(p) = \mu \ln^{1/k}\left(\frac{1}{1-p}\right) \quad \text{for } 0 \le p < 1. \tag{44}$$

For the Weibull distribution, the Normalizing and Centering Functions (12) are given by

$$b(\lambda t) = \mu \ln^{1/k}(\lambda t) \quad \text{and} \quad a(\lambda t) = \frac{\mu}{k} \ln^{\frac{1}{k}-1}(\lambda t). \tag{45}$$

The explicit approximations for the Weibull distribution are provided in Table 7.

9. Numerical Experiments

In this section, we numerically test the accuracy and robustness of our approximations and policies. Although our asymptotic results characterize the performance in

Table 7. Approximate Expressions Across Market Regimes Under a Weibull Customer Item Valuation Distribution

Approx. type	Small market $(q, \lambda t)$	Fluid $(q, \lambda t)$	Large market $(q, \lambda t)$
Optimal offline value	$\lambda t\mu\Gamma\left(1 + \frac{1}{k}\right)$	$\lambda t\mu\Gamma\left(1 + \frac{1}{k}, \ln\left(\frac{\lambda t}{q}\right)\right)$	$\mu q \ln^{1/k}\left(\frac{\lambda t}{e^{C_q/q}}\right)$
Optimal online value	$\lambda t\mu(ke)^{-1/k}$	$\mu q \ln^{1/k}\left(\frac{\lambda t}{q}\right)$	$\mu q \ln^{1/k}\left(\frac{(q/e)^q}{q!} \cdot \frac{\lambda t}{q}\right)$
Optimal price	$\mu k^{-1/k}$	$\mu \ln^{1/k}\left(\frac{\lambda t}{q}\right)$	$\mu \ln^{1/k}\left(\frac{\lambda t}{q}\right)$
Minimal regret	$\lambda t\mu\left(\Gamma\left(1 + \frac{1}{k}\right) - (ke)^{-1/k}\right)$	$\frac{\mu\lambda t}{k}\Gamma\left(\frac{1}{k}, \ln\left(\frac{\lambda t}{q}\right)\right)$	$\frac{\mu}{k}\ln\left(\frac{e^q q!}{e^{C_q}}\right)\ln^{\frac{1}{k}-1}(\lambda t)$

limiting regimes, it is essential to understand how these approximations behave in finite, operationally relevant settings. The experiments focus on the extent to which regime-specific approximations and the dynamic run-out rate policy retain their accuracy and effectiveness when system parameters are moderate, rather than extreme. First, in Subsection 9.1, we evaluate the three-regime value function approximations across a range of market sizes from Subsection 8.2. Next, in Subsection 9.2, we use the approximations from Subsection 8.2 to numerically determine the boundaries between the small market, fluid, and large market regimes in the online setting. In Subsection 9.3, we perform a similar procedure to determine the boundaries between the small market, fluid, and large market regimes in the offline setting. Finally, in Subsection 9.4, we assess the performance of the dynamic run-out rate policy from Subsection 8.4 and confirm its asymptotic optimality.

Throughout this section, we use Weibull-distributed item valuations with different shape parameters to examine the impact of different tail behavior on policy performance and approximation accuracy. The Weibull distribution with scale parameter $\mu > 0$ and shape parameter $k > 0$ is given

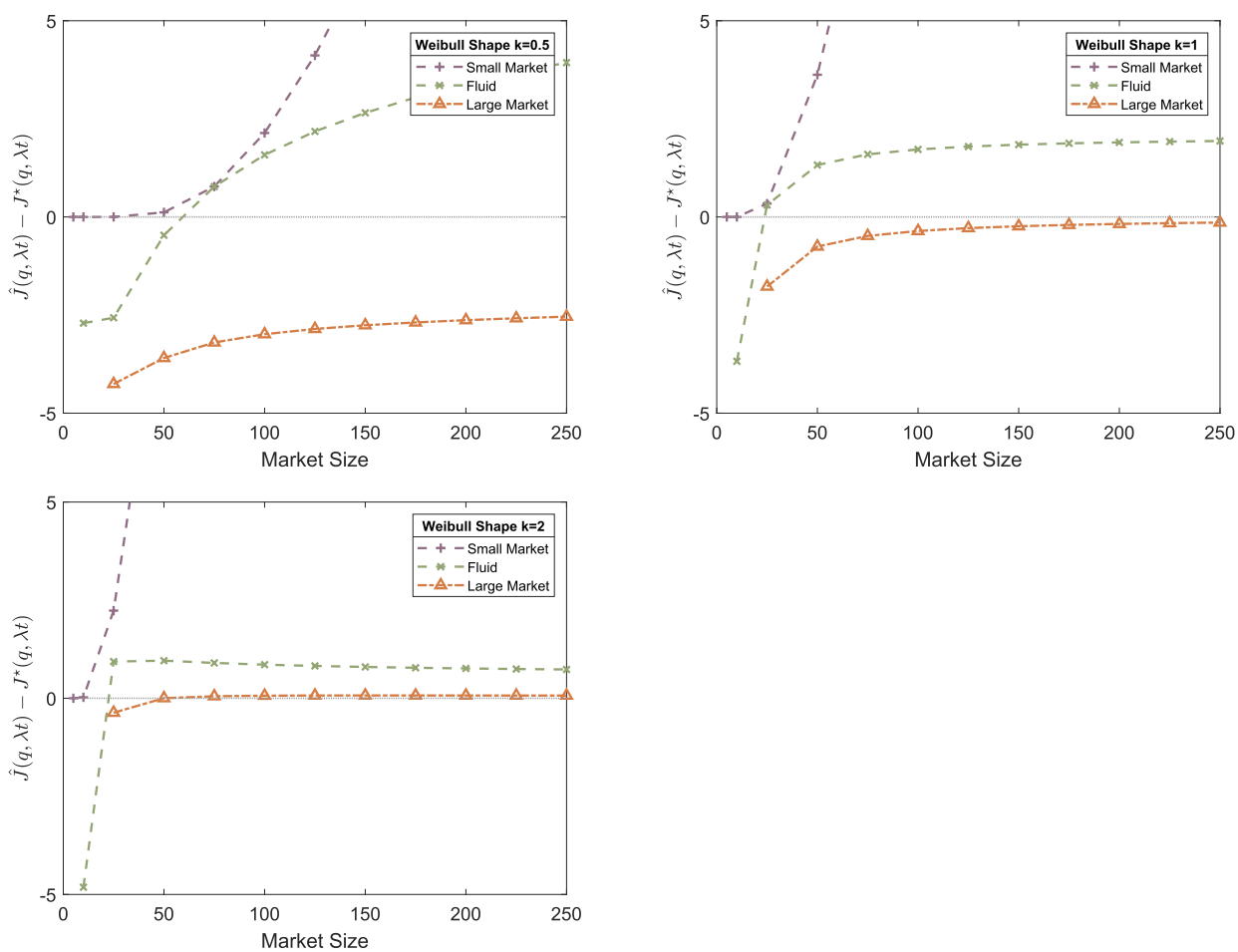
$$F(x) = 1 - \exp(-(x/\mu)^k) \text{ for } x \geq 0.$$

Recall that the Weibull distribution exhibits a decreasing hazard rate if $k < 1$ and an increasing hazard rate if $k > 1$. When $k = 1$, the Weibull distribution reduces to the exponential distribution that has a constant hazard rate. This makes the choice of Weibull-distributed item valuations a flexible one.

9.1. Accuracy of Optimal Value Function Approximations

We start by evaluating the accuracy of the small market, fluid, and large market approximations to the optimal online value function found in Table 4. Our approach is to compare each approximation to the true value of the optimal value function obtained by numerically solving the HJB Equations (2). The results are presented in Figure 5, where the initial inventory level q is kept fixed at 10 and the market size λt increases from 0 to 250. We consider three cases

Figure 5. (Color online) Accuracy of the Optimal Online Value Function Approximations for Item Valuations Distributed According to a Weibull Distribution (Initial Inventory $q = 10$)



corresponding to item valuations distributed according to a Weibull distribution with shape parameters $k = 1/2, 1$, and 2. In each case, the mean of the distribution is kept fixed at one by varying the scale parameter appropriately.

Note first that, as expected, for each value of the shape parameter, the small market approximation performs best for small values of λt, the fluid approximation is the most accurate for moderate values of λt, and the large market approximation performs best when λt is large. It is also apparent that the behavior of the hazard rate function of the item valuation distribution plays a significant role in our results in Figure 5. This corroborates our results from Subsection 9.4. Specifically, when the hazard rate function is decreasing (corresponding to the case of $k = 0.5$), the Large Market Approximation (27) provides a lower bound to the optimal value function with the bound tightening as λt increases. On the other hand, when the hazard rate function is increasing (corresponding to the case of $k = 2$), the Large Market Approximation (27) provides an upper bound to the optimal value function. The hazard rate function being constant (corresponding to $k = 1$) coincides with the norming function a being constant, and, therefore, the $o(a(\lambda t))$ term in (27) vanishes. Moreover, judging by Figure 5, it appears to converge to zero from below.

The fluid approximation to the optimal online value function found in Table 4 does not appear in Figure 5 to be as tight as the large market approximation. It actually turns out that when the item valuation distribution has a decreasing or constant hazard rate function ($k = 0.5$ or 1), the fluid approximation becomes looser as the market size λt increases. With an increasing hazard rate function ($k = 2$), the fluid approximation becomes tighter as λt grows large but at a slower rate than the large market approximation. As for the small market approximation, although it gives a tight approximation for small market sizes, it quickly diverges as the market size λt increases.

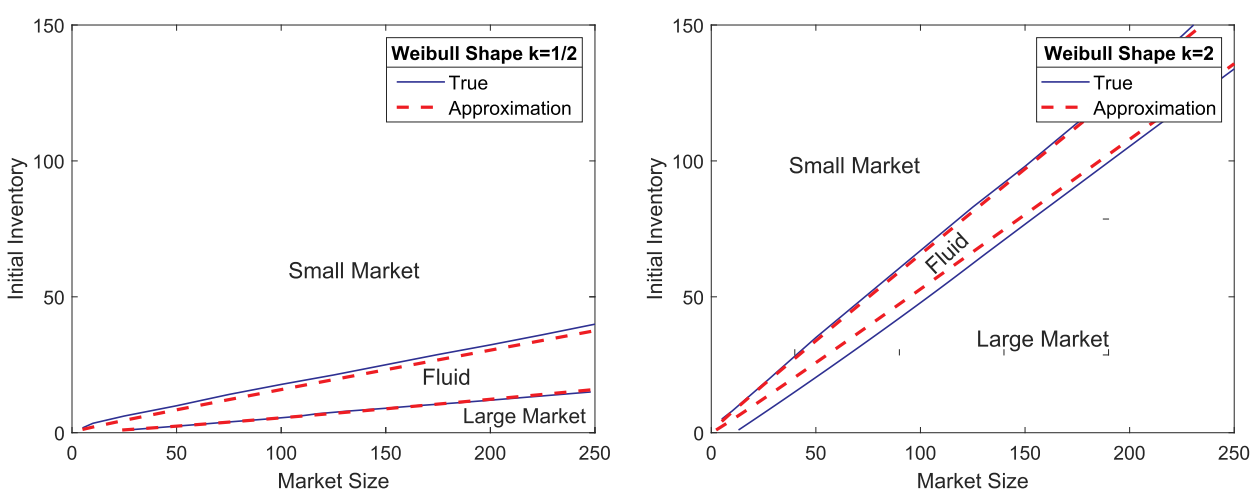
9.2. Online Regime Boundaries

We next study the boundaries between the three regimes in the online problem. For each pair $(q, \lambda t)$ of initial inventory levels and market sizes, we compare our optimal online value function approximations in Table 7 to the value of the optimal value function itself. Unfortunately, a closed-form expression for the optimal value function does not exist in the case of Weibull item valuation distributions. We therefore numerically solve the HJB Equations (2) to determine the optimal value function in this case.

Our results are presented in Figure 6, where we plot as a function of initial inventory level q and market size λt the regions where each of the optimal online value approximations of Table 7 is closest in absolute value to the optimal online value function. The plots correspond to customer item valuation distributions with shape parameters $k = 1/2$ and $k = 2$. In both cases, the scale parameter of the item valuation distribution has been adjusted so that its mean remains equal to one. Recall that the case of a shape parameter $k = 1$ corresponds to the exponential distribution studied in Section 5, and its plot is given in Figure 2.

The plots in Figure 6 for shape parameters $k = 1/2$ and 2 look similar to those in Figure 2 for the exponential case of $k = 1$. Each plot has three regions corresponding to where each asymptotic regime is optimal. The boundaries between the regions in each plot are approximately linear, corresponding to specific initial inventory-to-market size ratios. Moreover, the size of the fluid regime remains relatively consistent across different shape parameters k. However, higher values of k result in a delayed transition between regions

Figure 6. (Color online) Online Problem Boundaries Between the Three Asymptotic Regimes in the Case of Weibull Item Valuations



based on market size, which is indicated by the larger area of the small market regime.

The lack of a closed-form expression for the optimal value function makes it challenging to derive expressions for the boundaries between the small market, fluid, and large market regimes in the case of item valuations with a Weibull distribution. The following asymptotics that are plotted in Figure 6 do, however, appear to perform reasonably well. For the small market regime-fluid regime boundary, we have

$$\sigma^{-1}(\lambda t) = \frac{\lambda t}{e^{1/k}} + \gamma^{\star}\sqrt{\frac{2\lambda t}{e^{1/k}}} + o((\lambda t)^{1/2}) \text{ as } \lambda t \to \infty. \quad (46)$$

Next, for the fluid regime-large market regime boundary, we have

$$\tau(q) = qe^{1/k} + \frac{k^{1/2-1/k}}{(2\pi)^{k/4}} e^{1/k} q^{3/4} + o(q^{3/4}) \text{ as } q \to \infty. \quad (47)$$

9.3. Offline Regime Boundaries

We next study the boundaries between the three regimes in the offline problem. Again, we consider Weibull item valuation distributions. Our approach is similar to that in Subsection 9.2. For each pair $(q, \lambda t)$ of initial inventory levels and market sizes, we find the best (in terms of absolute value) offline approximation in Table 7. Note that the optimal offline value function is readily computable using (14) provided in Section 7.

As in Subsection 9.1, we consider item valuations with a Weibull distribution having shape parameters $k = 1/2, 1$, and 2. The scale parameter of each item valuation distribution is also adjusted so that its mean remains equal to one. Figure 7 provides our results and reveals some interesting insights into the offline problem approximations. In particular, note that for the case of item valuations that follow an exponential distribution ($k = 1$), the fluid approximation is dominated everywhere by either the small or large market approximation. On the other hand, for $k \neq 1$, we observe that the large market regions are small relative to the small market and fluid regions.

9.4. Performance of Asymptotically Optimal Policy

We complete the numerics section by comparing the value function J_p of the asymptotically optimal

Figure 7. (Color online) Offline Problem Boundaries Between the Three Asymptotic Regimes in the Case of Weibull Item Valuations

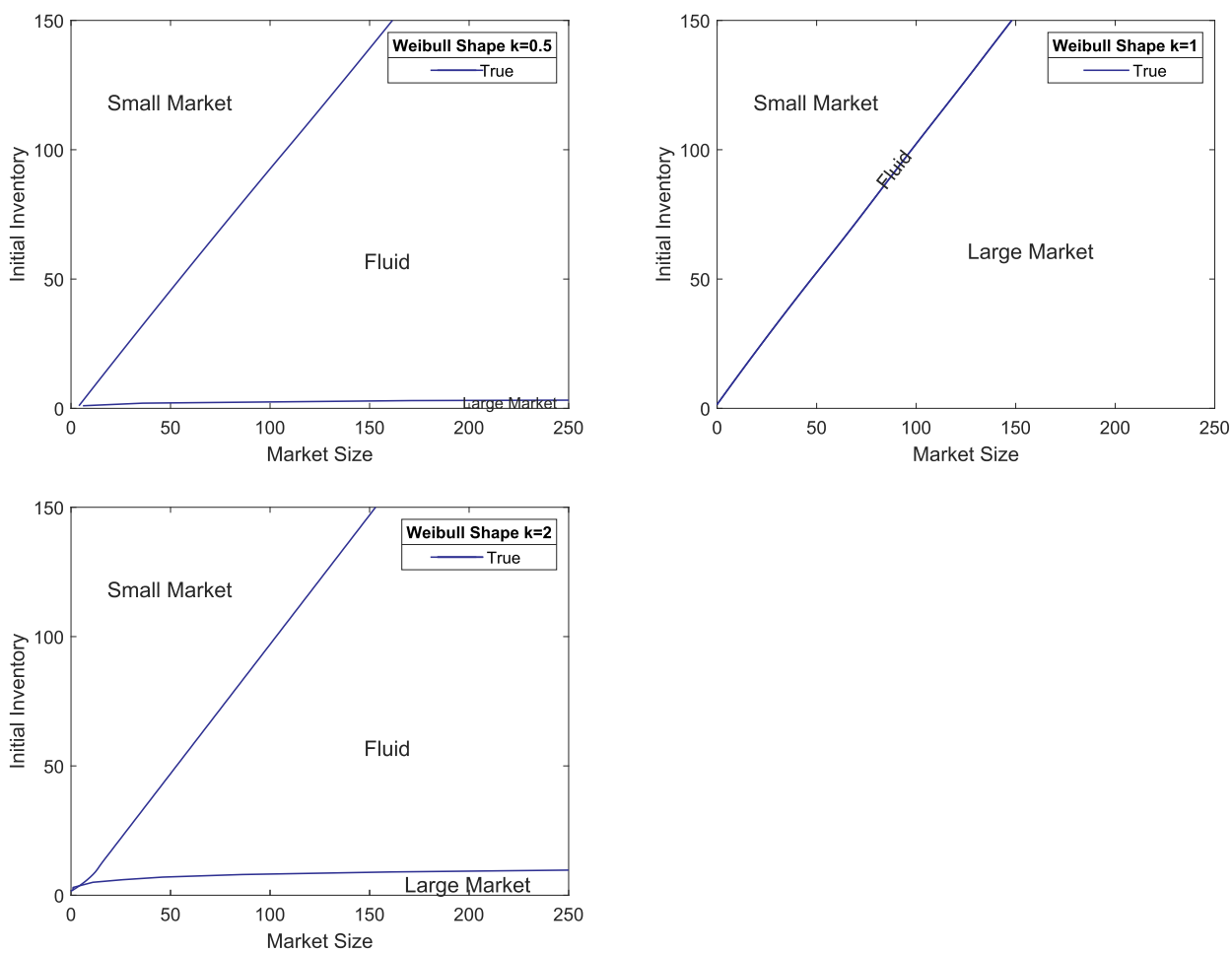
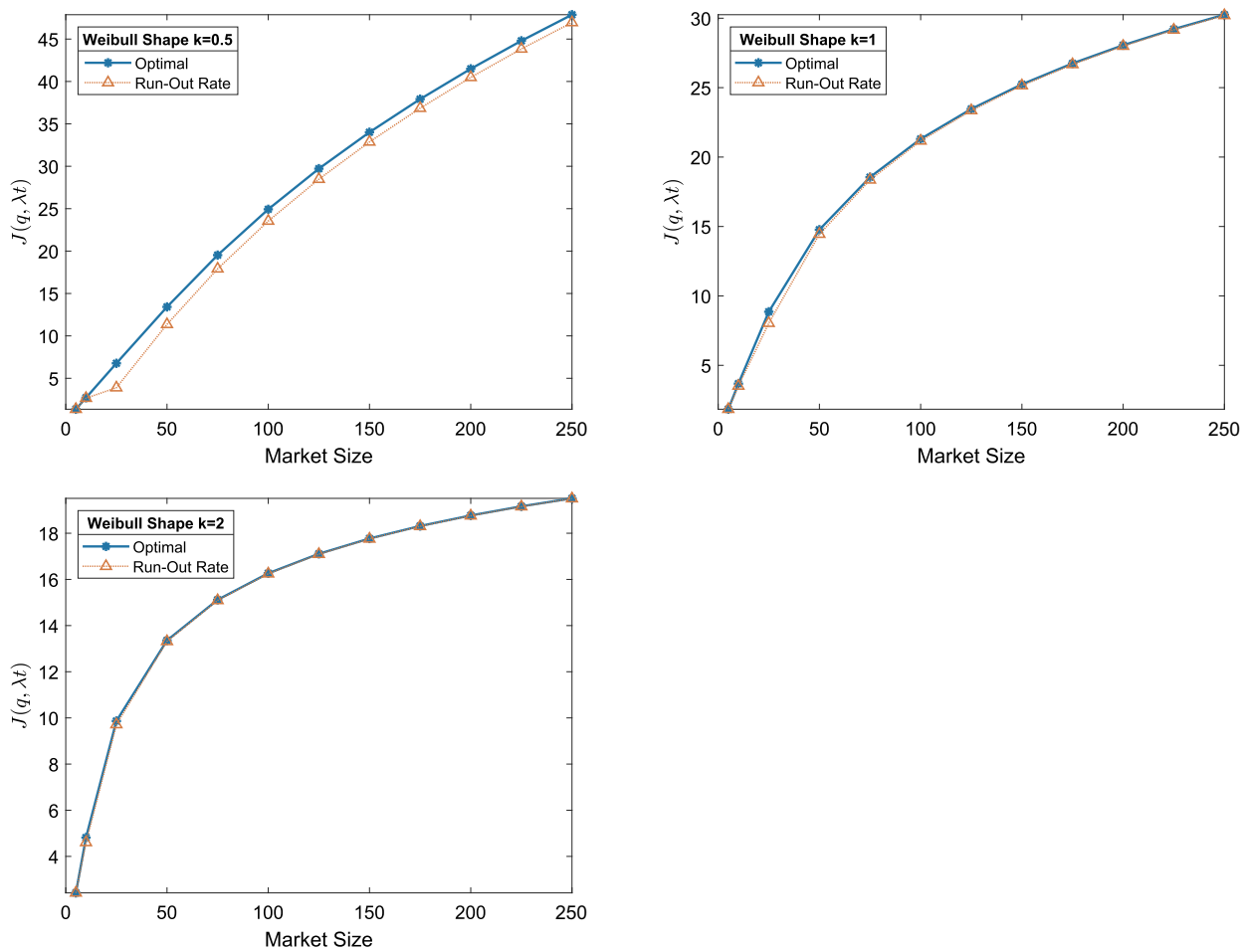


Figure 8. (Color online) Value Functions of the Optimal Pricing Policy and the Asymptotically Optimal Dynamic Run-out Rate Pricing Policy for Item Valuations Distributed According to a Weibull Distribution (Initial Inventory $q = 10$)



dynamic run-out rate pricing policy given by (39) from Subsection 8.4 against the optimal value function $J^\star$. Both of these value functions may be approximated by numerically solving the HJB Equations (2). The two value functions are plotted in Figure 8 assuming Weibull item valuation distributions with a mean of one and shape parameters $k = 1/2, 1$, and 2. The initial inventory level is held fixed at $q = 10$ units.

Recall from (38) of Subsection 8.4 that for a fixed initial inventory level $q \in \mathbb{N}_+$, the difference between the performance of any asymptotically optimal pricing policy and the optimal pricing policy is $o(a(\lambda t))$ as $\lambda t \to \infty$, where a is the Norming Function (12). Moreover, a may be expressed as a time change of the Mills ratio (or reciprocal hazard rate function) of the item valuation distribution F. The findings in Figure 8 bear this asymptotic out. For the most part, the dynamic run-out rate pricing policy achieves almost the same performance as the optimal pricing policy. However, if one inspects closely the plot corresponding to item valuations following a Weibull distribution with shape parameter $k = 1/2$, there is a gap between the expected revenue of the dynamic run-out rate pricing policy and that of the optimal pricing policy. This is not surprising because $k = 1/2$ corresponds to a decreasing hazard rate, which implies that the function $a(\lambda t)$ does not vanish.

10. Conclusion

This paper develops a framework for studying the single-item dynamic pricing problem based on three asymptotic regimes characterized by their initial inventory level-to-market size ratio. We provide approximations to both the offline and online optimal value function in each of these regimes. We also theoretically and numerically identify the boundaries between the small market, fluid, and large market regimes. The large market regime is new and asymptotically displays behaviors that deviate from the classical fluid regime. In particular, the first- and second-order approximations to the offline and online optimal value functions, as well as the minimal regret, deviate from those of the fluid regime. Despite these differences, we can still establish the robustness of a dynamic run-out

rate pricing policy in both the fluid and large market regimes.

There exist several directions for future research. One natural extension is to study the single-item dynamic pricing problem in the large market regime, where customer item valuation distributions lie in the remaining two extreme value domains of attraction—that is, the Weibull and Frechet domains. A second interesting direction for future research is to incorporate multi-item settings.

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Endnote

[1] We assume the existence of a null price p_∞ such that $1 - F(p_\infty) = 0$.

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